

Plough pan layer redirects infiltration via subsurface lateral flow limiting vertical hydrological connectivity in long-term tillage farmland

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ABSTRACT

Vertical hydrological connectivity in agricultural soils governs water infiltration, root-zone moisture replenishment, and groundwater recharge, all key to crop productivity and ecosystem resilience. Few field-scale studies have addressed how plough pans alter vertical hydrological connectivity by reshaping dominant soil water flow pathways and altering soil moisture dynamics in agricultural soils. The present study examines this issue in a wheat-maize rotation system using dual-depth double-ring infiltrometers combined with Brilliant Blue tracing. Soil property analysis revealed the pronounced physicochemical stratification. The plough pan (20–40 cm) exhibited significantly higher bulk density, lower porosity, and reduced organic matter and water-stable aggregate content compared to the overlying plough layer (0–20 cm) ($p < 0.01$). Staining patterns showed that the plough pans functioned as a hydraulic barrier that converted vertical infiltration into lateral flow upon the boundary of plough pan. Infiltration measurement demonstrated that initial lateral infiltration rate exceeded the initial vertical infiltration rate by 27%, and the steady-state lateral infiltration rate (115.24 mm h^{-1}) was approximately 25 times greater than the steady-state vertical infiltration rate (4.43 mm h^{-1}). The lateral cumulative infiltration was 259.9 mm and the vertical cumulative infiltration was 40.1 mm at a cumulative water input of 300 mm, with vertical infiltration accounting for only 13% of the total. After infiltration, nearly all water storage (96%) was retained in the plough layer (37.5 mm of 39.0 mm). These results highlight the plough pan as a hydrological barrier that redirects infiltration laterally and confines water storage to the plough layer, informing tillage and irrigation strategies to restore vertical hydrological connectivity and mitigate waterlogging in intensively cultivated farmlands.

1. Introduction

Vertical hydrological connectivity describes how easily water moves from the soil surface to deeper layers, primarily governed by soil structure (Merenlender and Matella, 2013; Zhang et al., 2021; Cui et al., 2022). This process governs the depth and rate of water movement through the soil profile (Conte and Ferro, 2018). In agricultural systems, vertical hydrological connectivity regulates nutrient delivery to the root zone, controls soil water storage capacity, and underpins ecosystem

services such as flood mitigation and aquifer replenishment (Dong et al., 2019; Wu et al., 2021). Consequently, vertical hydrological connectivity fundamentally determines both crop productivity and long-term ecosystem resilience through its control over soil water availability.

In intensively tilled agroecosystems, such as wheat-maize rotation systems, long-term tillage often creates a compacted plough pan below the tilling depth (Bertolino et al., 2010; Zhai et al., 2016). This dense layer, characterized by higher bulk density and lower porosity, reduces water infiltration rates (Martínez et al., 2008; Nawaz et al., 2013; Alaoui

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et al., 2018). Kim et al. (2010) reported an 8% increase in bulk density corresponded to a 69% reduction in saturated hydraulic conductivity, leading to more surface runoff. Singh and Hadda (2014) also reported lower infiltration rates with increased subsoil compaction. The plough pan redirects infiltrating water into subsurface lateral flow pathways. This has been evidenced by dye-tracing studies documenting substantial lateral movement in cultivated fields (Kasteel et al., 2007; Wang et al., 2011; Jarvis et al., 2016; Qiu et al., 2023).

The plough pan acts as a barrier, disrupting vertical infiltration and redirecting water laterally, altering soil hydrological patterns. This leads to agronomic and environmental consequences. By impeding vertical drainage, it causes prolonged near-saturation and surface ponding (Bertolino et al., 2010), which restrict gas exchange and create anaerobic conditions in the root zone. The compacted plough pan also confines root growth to shallow layers, elevating the risks of root rot, lodging, and crop mortality during waterlogging events (Ishaq et al., 2001; Singh and Hadda, 2014). At the field scale, these processes translate into reduced crop yields, accelerated surface runoff, and elevated greenhouse gas emissions from anaerobic soils (Holman et al., 2003; Doerner and Horn, 2006; Alaoui et al., 2018). The lateral flow that develops along the plough pan boundary redistributes water within the profile (Verbist et al., 2007; Cuffe et al., 2020; Ehrhardt and Gerke, 2023) and can carry dissolved nitrates, pesticides, and heavy metals toward nearby waterways, raising pollution risks and complicating water-quality management (Liang et al., 2008; Li et al., 2021; Chen et al., 2022). Quantifying how infiltration partitions between vertical and lateral pathways is therefore essential not only for predicting agrochemical transport directions and fluxes but also for refining field-scale hydrological models in agricultural fields, which have traditionally assumed purely vertical flow.

Many methods have been developed to study subsurface lateral infiltration, including time-lapse ground-penetrating radar, dye-stained area indices, Hydrus-based water-balance modelling, undisturbed soil monoliths instrumented with tension discs, and soil-temperature contrast methods. Using time-lapse ground-penetrating radar combined with controlled water application, Guo et al. (2014) detected infiltration-induced changes in dielectric permittivity and wave propagation. By integrating these anomalies across successive radargrams, they reconstructed a three-dimensional network of lateral preferential flow pathways within the soil profile. However, the method visualizes flow pathways without quantifying lateral infiltration volumes. Wang and Zhang (2017) characterized lateral flow with an index that combined the maximum dye migration distance with the equivalent radius of the stained area; however, this index yields only the relative proportion of lateral flow and cannot quantify its absolute amount. Wang et al. (2011) estimated lateral infiltration indirectly through a Hydrus-2D water balance in a peanut cropping system. Lateral infiltration was calculated as the residual after subtracting overland flow, actual evapotranspiration, canopy interception, and the change in soil water storage from precipitation. Because the soil water storage change was derived from Hydrus-2D simulation rather than direct measurement, the estimate was sensitive to uncertainties in the inversely calibrated van Genuchten parameters.

In the laboratory, Ehrhardt et al. (2022) separated lateral and vertical flow in an undisturbed soil monolith extracted across a horizon boundary. Tension discs were installed at both the base of the monolith and along the lateral horizon boundary, enabling lateral infiltration to be distinguished from total infiltration. However, the effective area of the lateral tension disc was relatively large compared with the monolith's cross-section, potentially disturbing the internal pressure field. Because infiltrating water alters soil temperature, Ehrhardt and Gerke (2023) detected subsurface lateral flow under field conditions by comparing soil temperature time series between a vertical lysimeter (which permits only vertical infiltration) and the surrounding field soil where water can move freely in all directions. However, the authors noted that this method was better suited for qualitative identification of

lateral infiltration; quantitative analysis still required cross-validation with measured soil moisture data.

Previous studies have investigated lateral infiltration either by characterizing its relative proportion, by measuring it on undisturbed soil monoliths in the laboratory, or by inferring it from contrasts in soil temperature responses. However, the quantitative separation of lateral and vertical infiltration under field conditions is a rarely explored advance. The double-ring infiltrometer, widely used owing to its simple design, ease of operation, and low cost, offers a practical foundation for this purpose. In agricultural fields, Liao et al. (2023, 2025) confirmed the occurrence of lateral flow using double-ring infiltrometers but did not quantify lateral infiltration and its extent relative to total infiltration. Therefore, this study employed two double-ring infiltrometers at contrasting insertion depths for measuring lateral infiltration in wheat-maize rotation fields with a plough pan. One was inserted to the depth of the plough layer to measure only vertical infiltration, and the other was inserted just below the soil surface to measure vertical and lateral infiltration. The lateral infiltration was obtained as the difference between the two. Brilliant Blue dye was applied to verify the presence of lateral flow. Therefore, the specific objectives are to: (1) characterize the physicochemical stratification associated with the plough pan across the soil profile; (2) quantify the partition of infiltration between vertical and lateral infiltration across the plough pan boundary and (3) examine the impact of this flow partitioning on post-infiltration soil water storage distribution. The results are expected to provide a mechanistic basis for understanding plough pan-mediated hydrological disruption and to inform strategies for enhancing water-use efficiency and reduce environmental trade-offs in farmlands management.

2. Materials and methods

2.1. Study area

This study was conducted in Yucheng county (115°55'E, 34°30'N), located in Henan Province on the Huang-Huai-Hai Plain, China (Fig. 1a). The plain is one of the major grain-producing regions in China. The study area is situated at an elevation of approximately 49 m above sea level, with flat alluvial terrain and gentle slopes of 1/6000–1/5000. The region experiences a temperate monsoon climate, with a mean annual temperature of 14.1°C and mean annual precipitation of 726.5 mm, of which approximately 58.1% is concentrated in July and August, making the area prone to summer waterlogging. The mean annual evaporation is 1151.1 mm from open water surfaces and 637.4 mm from land surfaces. Soils in the study area are classified as silt loam according to USDA soil texture classification, based on particle size analysis of the topsoil (0–20 cm), which contains approximately 0.6% clay, 66.6% silt, and 32.8% sand (Table 1). The experimental field has been continuously managed under a wheat-maize double-cropping rotation for over 25 years, with moldboard ploughing performed annually to a depth of 15–20 cm prior to winter wheat sowing (Zhai et al., 2016). The last ploughing before the experiment was carried out in October 2023. The field experiments were conducted in April 2024, when winter wheat was at the jointing stage. According to Yucheng National Basic Meteorological Observing Station, total rainfall in the 7 days before the experiments was less than 5 mm, and cumulative rainfall in the preceding 30 days was approximately 29.5 mm (Fig. 2).

2.2. Experimental design and measurement

Three experimental sites were selected within the study area, spaced approximately 1–5 km apart at comparable elevations, to maintain consistent land-use history. Previous research indicated the formation of a plough pan at a depth of 15–20 cm below the soil surface (Zhai et al., 2016). At each site, three plots (2 m × 6 m) were randomly established (Fig. 1b), yielding a total of nine plots. Herbicide was applied when wheat seedlings had developed 3–5 leaves. Surface vegetation was

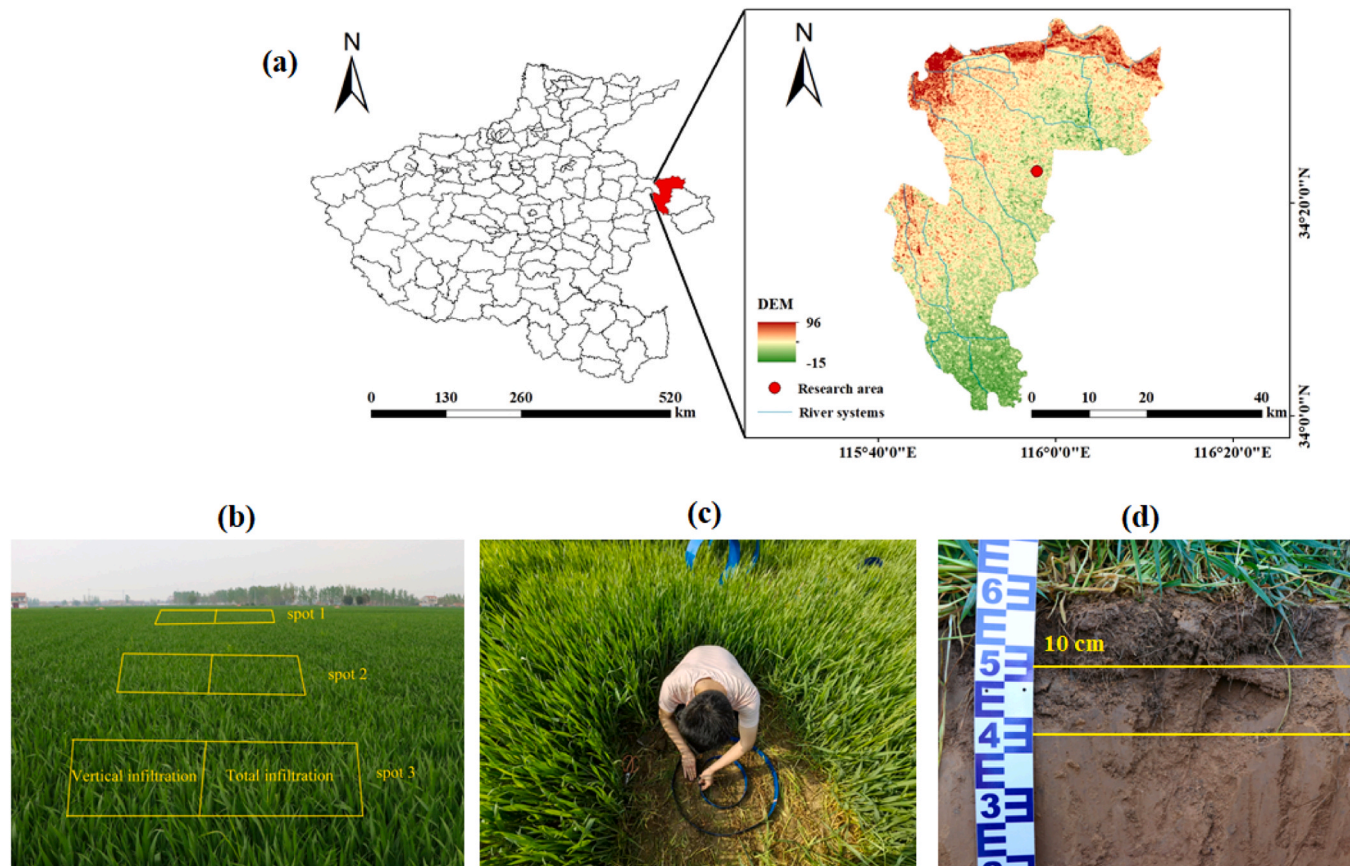


Fig. 1. The information and photos of the study area in (a) experimental area position (b) treatments in one site and (c) installation of a double ring infiltrometer (d) sampling location.

Table 1
Basic soil physicochemical properties of vertical soil profiles for the three experimental sites.

Depth (cm)	SWC (%)	BD (g cm^{-3})	SOM (g kg^{-1})	Clay (%)	Silt (%)	Sand (%)	SA (%)
0–10	$24.67 \pm 1.15\text{a}$	$1.28 \pm 0.08\text{b}$	$29.11 \pm 4.51\text{a}$	$0.67 \pm 0.23\text{a}$	$66.10 \pm 5.57\text{a}$	$33.23 \pm 5.78\text{a}$	$71.37 \pm 2.86\text{a}$
10–20	$26.82 \pm 1.73\text{ab}$	$1.41 \pm 0.10\text{b}$	$18.06 \pm 7.51\text{a}$	$0.66 \pm 0.42\text{b}$	$67.04 \pm 9.85\text{b}$	$32.30 \pm 1.16\text{b}$	$38.98 \pm 10.55\text{a}$
20–30	$22.29 \pm 1.32\text{a}$	$1.56 \pm 0.14\text{b}$	$10.57 \pm 1.46\text{a}$	$1.63 \pm 0.21\text{a}$	$76.18 \pm 3.29\text{a}$	$22.19 \pm 3.26\text{a}$	$37.76 \pm 8.15\text{a}$
30–40	$21.79 \pm 0.73\text{a}$	$1.52 \pm 0.04\text{a}$	$7.07 \pm 0.39\text{ab}$	$0.96 \pm 0.33\text{a}$	$68.44 \pm 10.21\text{a}$	$30.60 \pm 10.44\text{a}$	$24.34 \pm 16.24\text{a}$
40–50	$24.07 \pm 0.76\text{a}$	$1.49 \pm 0.02\text{a}$	$7.36 \pm 1.66\text{a}$	$1.41 \pm 0.59\text{a}$	$66.12 \pm 12.30\text{a}$	$32.47 \pm 12.77\text{a}$	$18.40 \pm 15.39\text{a}$
50–60	$24.16 \pm 0.92\text{a}$	$1.45 \pm 0.10\text{a}$	$5.24 \pm 0.92\text{a}$	$0.98 \pm 0.41\text{a}$	$61.8 \pm 14.09\text{a}$	$37.22 \pm 6.92\text{a}$	$17.49 \pm 22.21\text{a}$
60–70	$24.58 \pm 3.51\text{a}$	$1.46 \pm 0.05\text{a}$	$6.58 \pm 0.91\text{a}$	$0.95 \pm 0.65\text{a}$	$61.46 \pm 11.44\text{a}$	$37.59 \pm 14.89\text{a}$	$10.90 \pm 11.46\text{a}$

Note: SWC is soil water content before infiltration experiments, %; BD is bulk density, g cm^{-3} ; SOM is soil organic matter, g kg^{-1} ; SA is soil water stable aggregates ($>0.25\text{ mm}$), %.

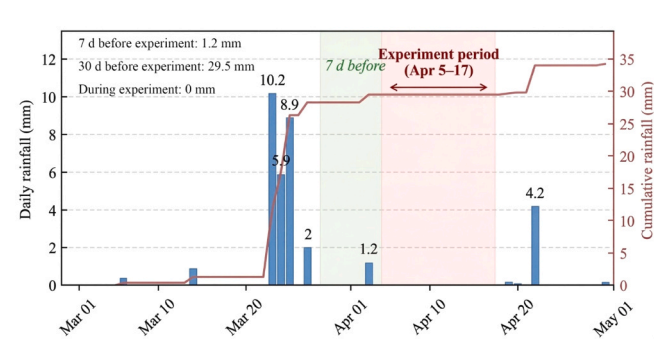


Fig. 2. Daily rainfall and cumulative rainfall at the study area during March–April 2024. The green shading indicates the 7-day antecedent period prior to the experiment, and the red shading indicates the experiment period (April 5–17).

predominantly wheat (Fig. 1c). Prior to the experiments, wheat plants were carefully cut close to the soil surface using a sickle, and surface litter was removed to minimize disturbance to the soil.

At each plot, two infiltration tests were conducted concurrently to separately quantify total infiltration (vertical + lateral) and vertical-only infiltration. Both infiltrometers consisted of an inner ring (30 cm diameter) and an outer ring (60 cm diameter), constructed from 2 mm thick stainless steel with sharpened lower edges to facilitate insertion into the soil. The two infiltrometers differed only in ring height and installation depth, which determined whether lateral flow was permitted or excluded. The shallow infiltrometer (ring height: 8 cm) was inserted 2–3 cm into the soil, leaving 5–6 cm above the surface (Fig. 3a). At this minimal insertion depth, water infiltrating through the inner ring was free to move both vertically and laterally within the plough layer, allowing the measurement of total infiltration. The deep infiltrometer (ring height: 30 cm) was inserted 15–20 cm into the soil, penetrating

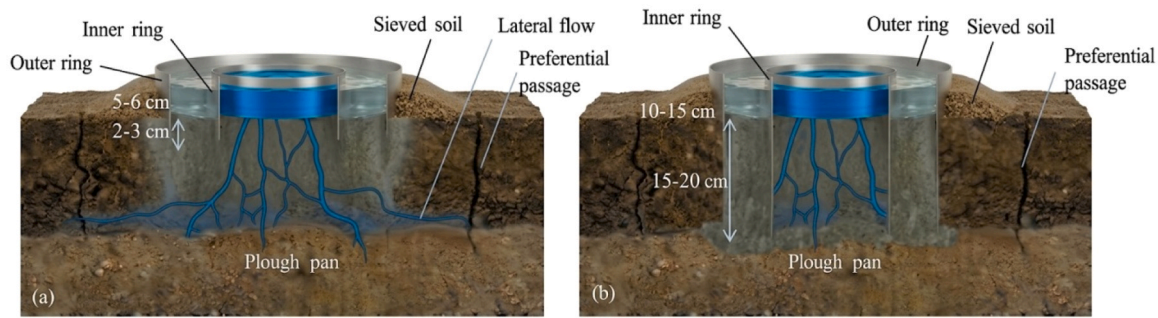


Fig. 3. Schematic cross-sections of the dual-depth double-ring infiltrometer installation. (a) Shallow installation (ring height: 8 cm; inserted 2–3 cm, with 5–6 cm above the soil surface) for measuring total infiltration (vertical + lateral infiltration). (b) Deep installation (ring height: 30 cm; inserted 15–20 cm, with 10–15 cm above the soil surface) for measuring vertical infiltration only.

through the plough layer and reaching the upper boundary of the plough pan (Fig. 3b). At this depth, the ring walls physically blocked lateral water movement within the plough layer, so that only vertical infiltration was measured. After installation, sieved soil (passed through a 2 mm mesh) collected from the experimental area was carefully packed around the outer rings to seal any gaps between the ring walls and the surrounding soil, preventing edge flow. The graduated ruler on the inner rings was used to record water level decline over time for calculating infiltration rates. A ruler attached to each outer ring was used to maintain equal water levels between the two rings throughout the experiment, ensuring that the hydraulic head in the outer ring matched that in the inner ring and thereby minimizing edge flow at the inner ring boundary.

Based on the dual-depth infiltrometer design described above, lateral infiltration through the plough layer was calculated as the difference between total infiltration and vertical infiltration by the following equation:

$$I_{\text{lateral}} = I_{\text{shallow}} - I_{\text{deep}} \quad (1)$$

where I_{shallow} is the infiltration measured by the shallow infiltrometer (inserted 2–3 cm), which captures both vertical and lateral infiltration. I_{deep} is the infiltration measured by the deep infiltrometer (inserted 15–20 cm), which captures vertical infiltration only. Eq. (1) is applied to both infiltration rates (mm h^{-1}) and cumulative infiltration (mm). The lateral infiltration rate (LIR) and lateral cumulative infiltration (LCI) are calculated as:

$$\text{LIR} = \text{TIR} - \text{VIR} \quad (2)$$

$$\text{LCI} = \text{TCI} - \text{VCI} \quad (3)$$

where TIR (mm h^{-1}) and TCI (mm) are the total infiltration rate and total cumulative infiltration measured by the shallow infiltrometer, respectively. VIR (mm h^{-1}) and VCI (mm) are the vertical infiltration rate and vertical cumulative infiltration measured by the deep infiltrometer, respectively. LIR (mm h^{-1}) and LCI (mm) are the lateral infiltration rate and lateral cumulative infiltration derived by subtraction.

A Brilliant Blue dye solution (4 g L^{-1} ; Beilian, Tianjin, China) was prepared before the experiment and added to the inner ring of both infiltrometers at the onset of each test. Brilliant Blue is a non-toxic, food-grade dye widely used in soil hydrological studies for its high visibility against soil matrices and low sorption to soil particles (Flury et al., 1994; Pinos et al., 2023). The outer ring was filled with clear water to the same level as the inner ring. During each test, the time required for each 0.5 cm (during the rapid-infiltration phase) or 1.0 cm (after the rate stabilized) decline of the inner-ring water level was recorded with a stopwatch. The infiltration rate was calculated with Eq. (4). The experiment continued until the infiltration rate reached a steady state. The steady-state infiltration rate was defined as the infiltration rate at which three consecutive measurements showed no appreciable change.

The initial infiltration rate was defined as the mean infiltration rate during the first 5 min of the experiment. Initial vertical infiltration rate is abbreviated as IVIR. Initial lateral infiltration rate is abbreviated as ILIR. Steady-state vertical infiltration rate is abbreviated as SVIR. Steady-state lateral infiltration rate is abbreviated as SLIR.

$$i = \frac{\Delta I}{\Delta t} \quad (4)$$

where i is infiltration rate, mm h^{-1} , ΔI (mm) is the change in water level within the inner ring over Δt (h).

2.3. Soil profile sampling

Approximately 24 h after the infiltration test, two parallel vertical soil profiles were exposed at each plot by excavating a trench passing through the center of the double-ring infiltrometer. Each profile was 200 cm wide and 100 cm deep. During excavation, earthworms and other burrowing animals were observed in the plough layer.

Loose surface soil particles were carefully removed using a small brush and a handheld fan. After clearing, photographs were taken under a shade cloth using a Canon digital camera. Profiles were then photographed using a Canon digital camera under a black diffusing cloth that provided uniform shading and indirect lighting. All photographs were taken by the same operator under identical conditions to minimize inter-observer variability. The photographs were processed in Photoshop CC 2017 to correct geometric distortion and adjust hue, saturation, and brightness. The images were then converted to binary (black and white) format. Stained areas were quantified from the binary images using the Tabulated Area tool in ArcGIS 10.2, following Cui et al. (2022). After photography, soil samples were collected from the center of the inner-ring footprint at seven depth intervals (0–10 cm, 10–20 cm, 20–30 cm, 30–40 cm, 40–50 cm, 50–60 cm, and 60–70 cm). At each depth, three sub-samples were combined into a single composite sample. Soil samples were placed in sealed aluminum tins, weighed, and immediately oven-dried at 105°C for 24 h to determine the post-infiltration soil water content (SWC).

Prior to the infiltration tests, undisturbed and disturbed soil samples were collected at two sampling locations within each plot at the same seven depth intervals. At each location, one undisturbed core sample per depth was collected with a stainless steel cutting ring (100 cm^3 ; 50.46 mm diameter $\times$ 50 mm height) for bulk density (BD) determination and transported in a rigid container to preserve structure. For soil organic matter (SOM), and particle size distribution, three subsamples were taken with a soil auger, homogenized into a single composite sample, sealed in a polyethylene bag, and stored in a cooler during transport. For water-stable aggregate (SA) analysis, intact soil blocks ($200 \text{ mm} \times 60 \text{ mm} \times 40 \text{ mm}$) were collected in square aluminum boxes and handled with care to preserve aggregate structure. All samples were transported to the laboratory within 24 h. This yielded 18 independent

measurements (three sites $\times$ three replicate plots $\times$ two profiles) per depth interval for each soil property.

2.4. Soil properties measurements

Soil texture (clay, silt and sand fractions) was determined by laser diffraction using a Mastersizer 2000 particle size analyzer (Malvern Panalytical Ltd., Malvern, Worcestershire, UK). The Soil organic matter (SOM) content was measured by the potassium dichromate oxidation method (Dong et al., 2022). Water-stable aggregate (SA) was determined by the wet-sieving method (Liao et al., 2023). Bulk density (BD) was determined from the oven-dried mass (105°C, 24 h) and the known volume of each undisturbed core sample. Total porosity (TP) was calculated as Eq. (5).

Total porosity (TP) was calculated by Eq. (5):

$$\varphi = 1 - \frac{\rho_b}{\rho_s} \times 100\% \quad (5)$$

where φ is total porosity (TP), ρ_b is soil bulk density (g cm^{-3}); ρ_s is the soil particle density, taken as 2.65 g cm^{-3} for mineral soils (Hillel, 2003).

The gravimetric soil water content (SWC) was determined for all samples collected both before and 24 h after the infiltration experiments by oven drying at 105 °C for 24 h. SWC was calculated by Eq. (6):

$$\theta_g = \frac{M_{\text{wet}} - M_{\text{dry}}}{M_{\text{dry}}} \times 100\% \quad (6)$$

where θ_g is gravimetric soil water content (SWC, %). M_{wet} is the gravimetric value of wet soil (g), and M_{dry} is the gravimetric value of dry soil (g).

Soil water storage (SWS, mm) is the thickness of the water layer formed by all the water in a certain soil layer (Milly, 1994) and was computed for each depth by the following formula:

$$W = \rho_b \times \theta_g \times H \quad (7)$$

where W is soil water storage (SWS, mm) over a layer of thickness H (mm). θ_g is the gravimetric water content (g g^{-1}), and ρ_b (g cm^{-3}) is the bulk density.

The increase in soil water storage (ISWS, mm) on the vertical profiles after infiltration was calculated with the following formula (Eq. 8) (Liao et al., 2023):

$$\Delta W = W_1 - W_0 \quad (8)$$

where ΔW is the increase in soil water storage (ISWS, mm), and W_0 (mm) and W_1 (mm) are the storage before and after the experiment, respectively.

2.5. Statistical analysis

Statistical analyses were performed using Python 3.11. To evaluate whether soil property differences between the plough layer and the plough pan were statistically significant, one-way analysis of variance (ANOVA) was applied to each soil property, with the two layers as grouping factors. To verify the robustness of the parametric ANOVA results without assuming normality, permutation tests were also conducted for each property, in which the observed difference in group means was compared against the distribution of differences generated by random reassignment of observations between layers. Additionally, among-site comparisons were performed using one-way ANOVA at each depth interval across the three experimental sites ($n = 6$ per site per depth) to assess whether the observed vertical stratification was spatially consistent or attributable to site-specific variability. Significance was reported at three thresholds, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

3. Results

3.1. Physicochemical stratification of the plough pan

Among the three experimental sites, no significant differences were detected for BD, TP, SWC, SA, clay, silt or sand at any depth interval (one-way ANOVA, all $p > 0.05$) (Fig. 4). Differences were observed for SOM at 10–20 cm ($p < 0.05$) depth among three sites, with Site 2 ($25.46 \pm 7.79 \text{ g kg}^{-1}$) exhibiting higher values than Site 1 ($13.77 \pm 1.57 \text{ g kg}^{-1}$) and Site 3 ($14.95 \pm 6.37 \text{ g kg}^{-1}$) (Fig. 4).

Soil bulk density (BD) increased from approximately 1.3 g cm^{-3} at the depth of 0–20 cm to about 1.5 g cm^{-3} at a depth of 20–40 cm, then slightly decreased and stabilized at around 1.4 g cm^{-3} below 40 cm ($p < 0.01$) (Table 1; Fig. 5a, b). Clear vertical stratification was observed between the 0–20 cm and 20–40 cm layers ($p < 0.01$) (Fig. 5). Soil water content (SWC) decreased from approximately 25.7% at a depth of 0–20 cm to about 22.0% at a depth of 20–40 cm (Fig. 5c) ($p < 0.01$). Soil total porosity (TP) decreased from approximately 49.2% at a depth of 0–20 cm to about 41.9% at a depth of 20–40 cm ($p < 0.01$) (Fig. 5d). SOM content declined sharply with depth, from 23.6 g kg^{-1} at a depth of 0–20 cm to 8.8 g kg^{-1} at a depth of 20–40 cm ($p < 0.01$) (Fig. 5e). The proportion of water-stable aggregates ($>0.25 \text{ mm}$) showed a similar decreasing trend, declining from 55.1% at a depth of 0–20 cm to 31.1% at a depth of 20–40 cm ($p < 0.01$) (Fig. 5f). The proportion of clay content increased from 0.7% at a depth of 0–20 cm to about 1.3% at a depth of 20–40 cm ($p < 0.01$) (Fig. 5g). The proportion of sand content decreased from approximately 32.8% at a depth of 0–20 cm to about 26.4% at a depth of 20–40 cm ($p < 0.01$) (Fig. 5h). This vertical stratification of soil physicochemical properties defined the plough layer (0–20 cm) and plough pan (20–40 cm).

3.2. Subsurface lateral flow dynamics

Dye-staining patterns in the excavated profiles revealed contrasting flow pathways between the two infiltrometer tests. In both, the dye terminated abruptly at the upper boundary of the plough pan, with a maximum penetration depth reaching 19.3 cm and no staining detected below the plough pan (Fig. 6a, b). In the deep infiltrometer profiles, the stained area decreased progressively through the plough layer and was nearly zero at 15–20 cm depth ($< 0.5 \text{ cm}^2$, below the practical resolution of the binary image analysis method) (Fig. 6c). In the shallow infiltrometer profiles, the stained area first decreased from the surface to approximately 10 cm depth, then increased between 10 and 15 cm depth, before declining sharply to near zero at 20 cm depth (Fig. 6d). This “decreasing-increasing-decreasing” pattern indicated that infiltrating water, upon encountering the plough pan, was redirected laterally, expanding the stained area above the pan boundary.

Both lateral and vertical infiltration rates exhibited time-dependent declines (Fig. 7a). VIR decreased by 78% within the first 5 min (from 428.4 mm h^{-1} to 95.1 mm h^{-1}), before stabilizing at 4.4 mm h^{-1} . The initial lateral infiltration rate (ILIR: 545.51 mm h^{-1}) was 27% higher than the initial vertical infiltration rate (IVIR: 428.36 mm h^{-1}) (Fig. 7b). The steady lateral infiltration rate (SLIR: 115.24 mm h^{-1}) was 25 times greater than the steady vertical infiltration rate (SVIR: 4.43 mm h^{-1}) (Fig. 7c). With increasing cumulative water input, the difference between vertical and lateral infiltration became increasingly pronounced (Fig. 7d). At 50 mm cumulative water input, vertical cumulative infiltration (VCI) was 19.8 mm and lateral cumulative infiltration (LCI) was 30.2 mm, with LCI accounting for 60% of total infiltration. As cumulative water input increased to 100, 200, and 300 mm, LCI increased to 76.3, 165.5, and 259.9 mm, respectively, while VCI increased to 23.7, 34.5, and 40.1 mm. At 300 mm cumulative water input, lateral infiltration accounted for 87% of TCI.

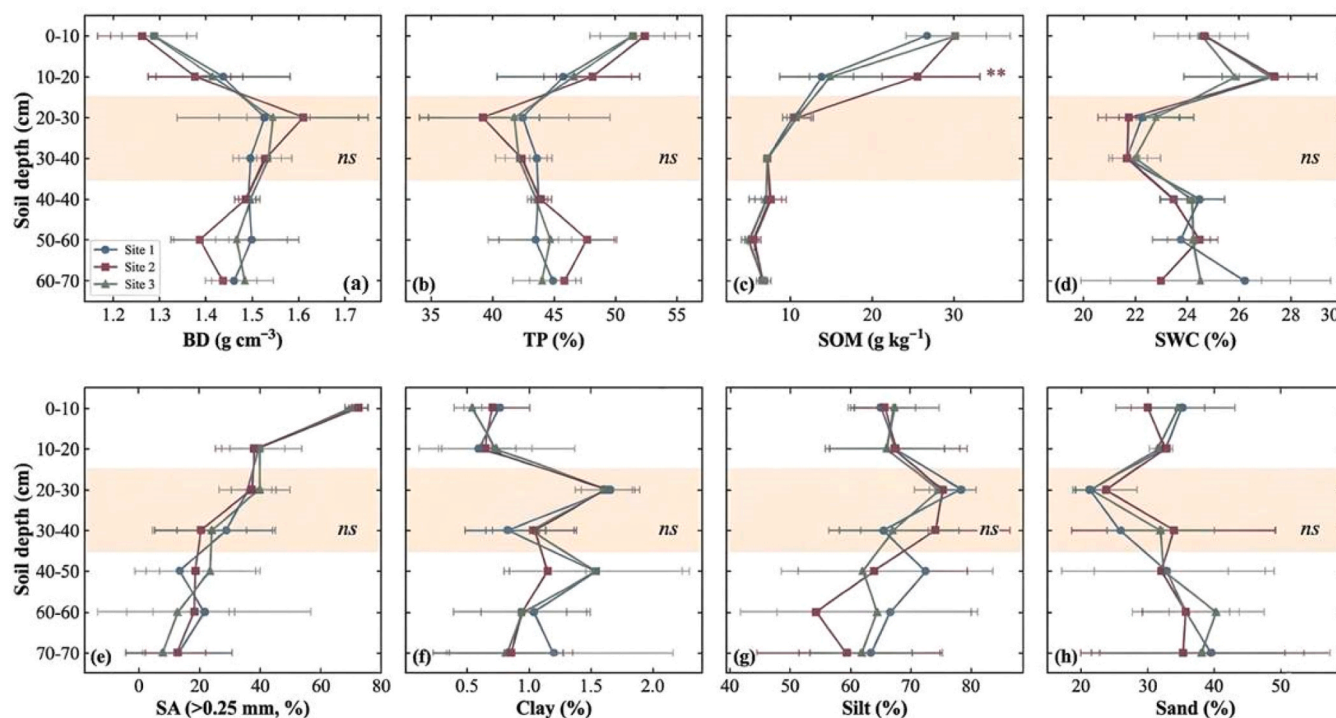


Fig. 4. Depth profiles of soil physicochemical properties at the three experimental sites: (a) bulk density (BD), (b) total porosity (TP), (c) soil organic matter (SOM), (d) soil water content (SWC), (e) water-stable aggregates (SA > 0.25 mm), (f) clay content, (g) silt content, and (h) sand content. Values represent means $\pm$ standard deviation ($n = 6$ per site per depth). The orange shading indicates the plough pan (20–40 cm). Significance markers on the right side of each panel indicate among-site differences at each depth interval (one-way ANOVA): * $p < 0.05$, ns = not significant.

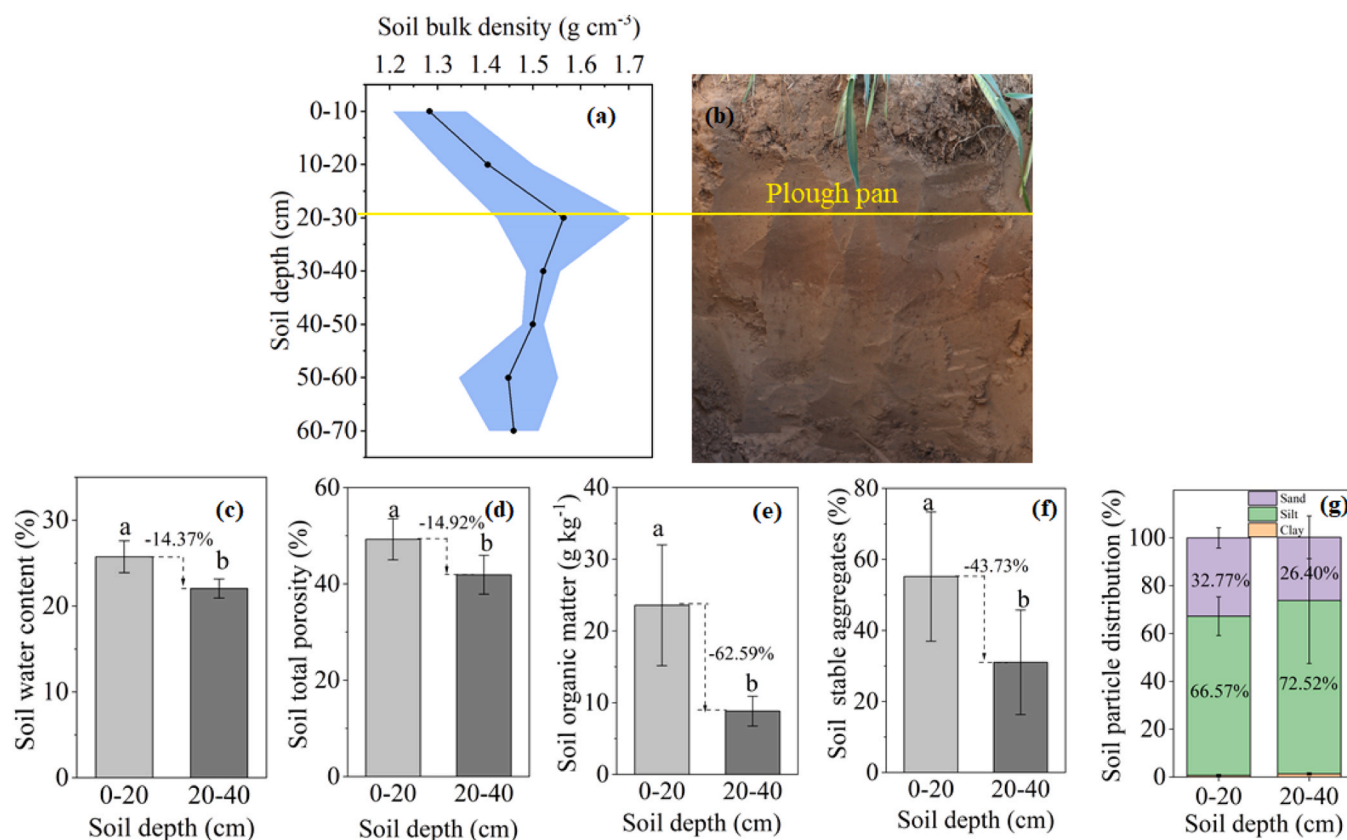


Fig. 5. Position of plough pan in (a) and (b) and comparison of physicochemical properties between plough pan and tillage layer in (c) soil water content, (d) soil total porosities, (e) soil organic matter, (f) soil water stable aggregates and (g) soil particle content.

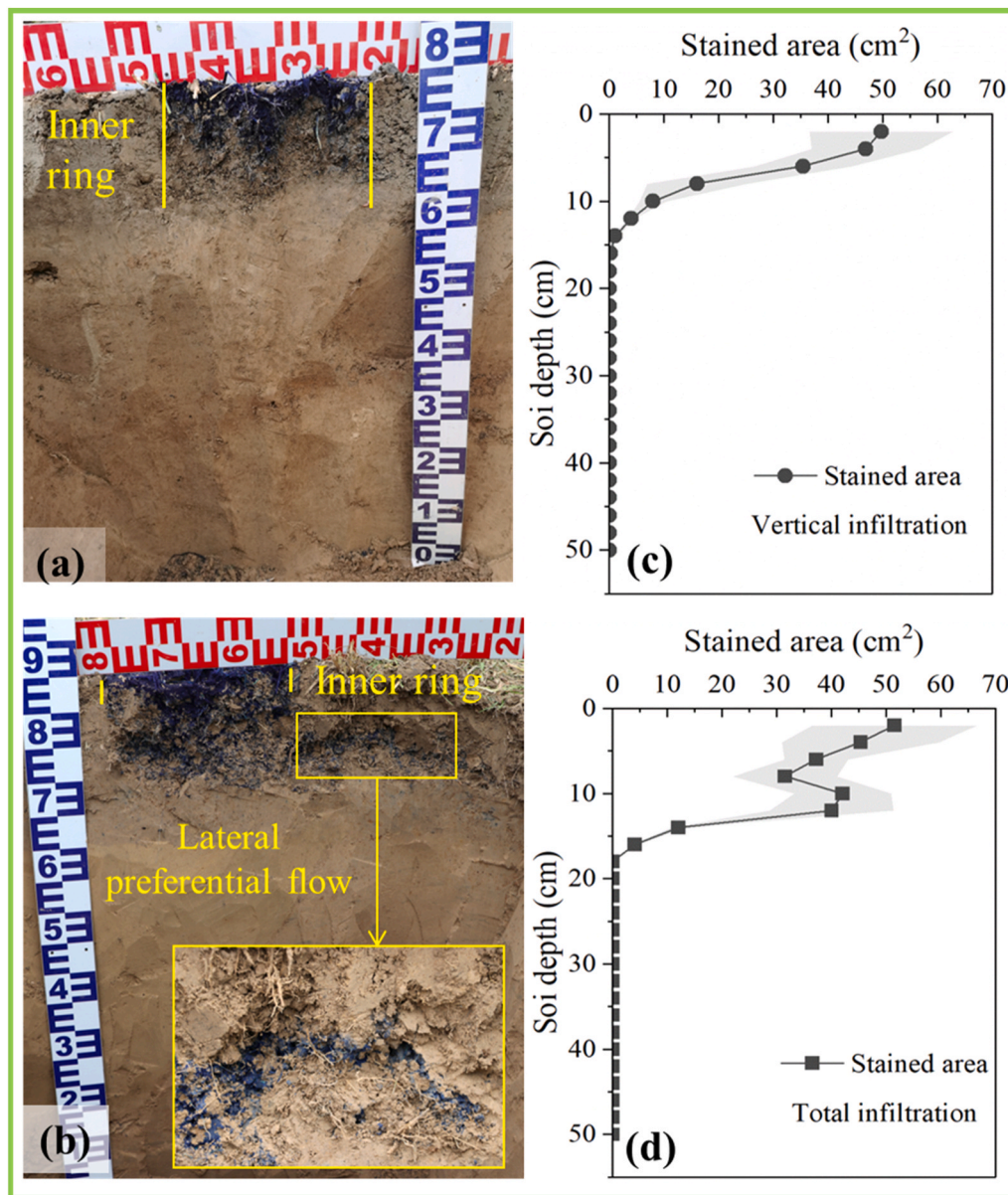


Fig. 6. Comparison of staining characteristics on the vertical soil profile between vertical infiltration (a) and total infiltration (b); the changing process of stained area with soil depth in vertical infiltration (c) and total infiltration (d).

3.3. Plough pan-mediated partitioning of soil water storage

Post-infiltration soil moisture profiles showed pronounced vertical stratification, consistent with the plough pan acting as a hydrological barrier (Fig. 8a, b). In the deep infiltrometer test (vertical infiltration only), SWC increased from 24.7% to 39.3% in the 0–10 cm layer and from 26.8% to 38.9% in the 10–20 cm layer after the infiltration experiment. In contrast, SWC in the plough pan (20–40 cm) rose only marginally, from 22.0% to 23.1%, and showed no measurable change below 40 cm (Fig. 8a). The increase in soil water storage (ISWS) showed a similarly concentrated distribution (Fig. 8b). The plough layer (0–20 cm) functioned as the dominant soil moisture reservoir, retaining 37.5 mm of the 39.0 mm total ISWS gained across the 0–70 cm profile, or 96% of the whole-profile increment (Fig. 8b).

4. Discussion

4.1. Physical stratification disruption in the plough pan

The pre-infiltration SWC at 60–70 cm depth showed greater variability than the shallower layers (Table 1). This variability may be attributable to spatial differences in groundwater table depth, which ranged from approximately 1.0–2.5 m below the soil surface across the experimental area (Zhang and Yan, 2021). Plots overlying a shallower water table received stronger upward capillary supply and tended to maintain higher SWC, whereas those above a deeper water table remained comparatively dry. No such effect was apparent in the 0–40 cm layers, where antecedent rainfall was the dominant control on SWC. The significant among-site difference in SOM at 10–20 cm depth ($p < 0.05$; Fig. 4c) was likely an artifact of sampling precision at this transitional zone between the plough layer and the plough pan. At this depth, SOM changes from 29.1 g kg⁻¹ (0–10 cm) to 10.1 g kg⁻¹ (20–30 cm) (Fig. 4c), and minor variations in the actual

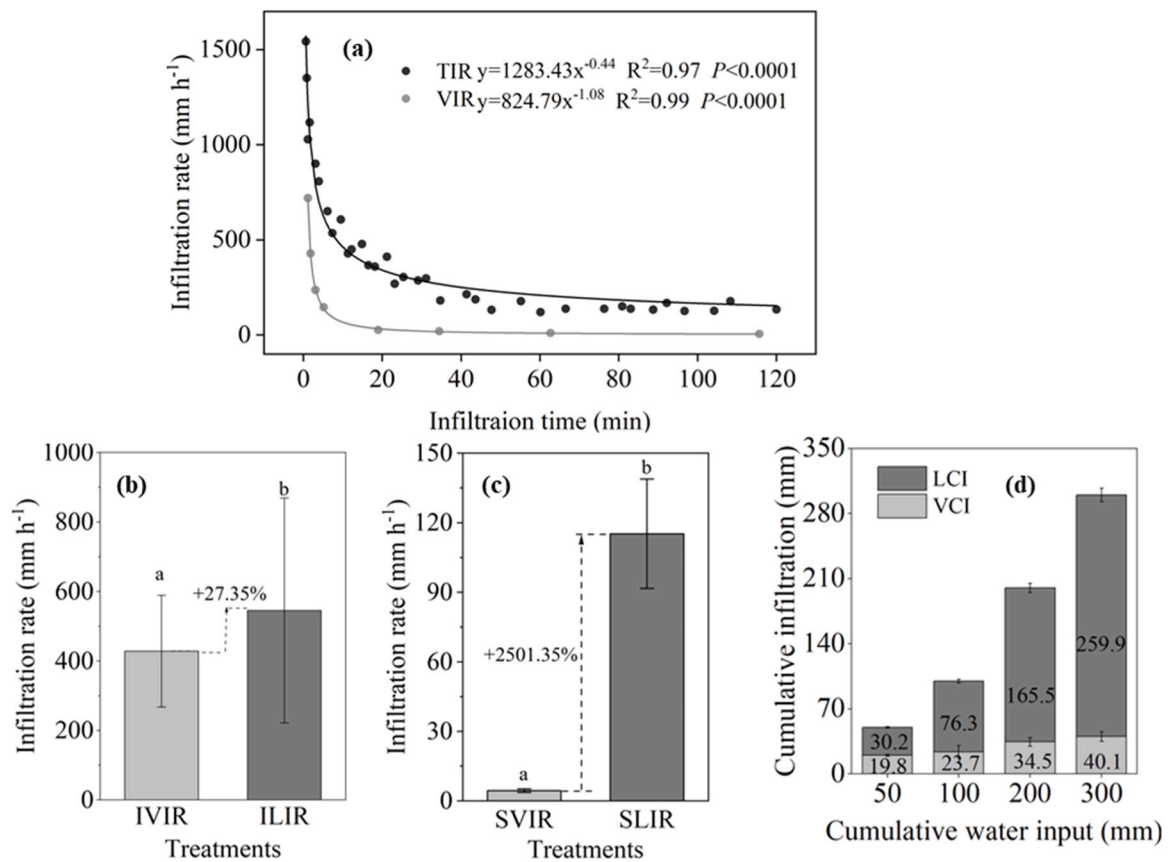


Fig. 7. Infiltration characteristics measured by dual-depth double-ring infiltrometers. (a) Total infiltration rate (TIR) and vertical infiltration rate (VIR) as functions of time, with fitted power-function equations. (b) Comparison of initial vertical infiltration rate (IVIR) and initial lateral infiltration rate (ILIR). (c) Comparison of steady-state vertical infiltration rate (SVIR) and steady-state lateral infiltration rate (SLIR). (d) Vertical cumulative infiltration (VCI) and lateral cumulative infiltration (LCI) at cumulative water inputs of 50, 100, 200, and 300 mm.

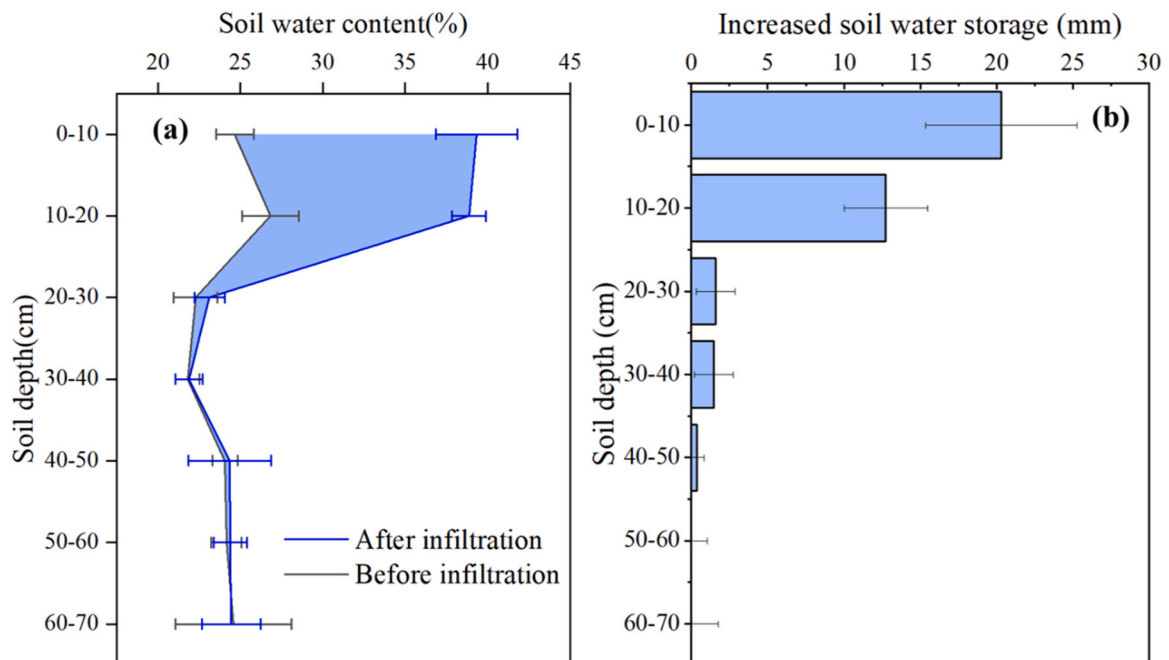


Fig. 8. Comparison of soil water content at different depth between before and after infiltration in (a); increased soil water storage in soil layers in different depths in (b).

sampling depth (1–2 cm) may introduce substantial differences in measured values. Except for SOM at 10–20 cm, there were no significant differences in the physicochemical properties of each soil layer among the three experimental sites.

Long-term mechanical tillage compacts the soil below the plough layer, and progressively eroding vertical hydrological connectivity. Repeated moldboard ploughing at a consistent depth of 15–20 cm over approximately 25 years has produced a well-defined plough pan in the 20–40 cm layer. These findings are consistent with the compaction-induced stratification reported at the same study area by [Zhai et al. \(2016\)](#). The compacted plough pan restricts vertical root growth, prevents crop roots from accessing deeper water resources ([Shu et al., 2014](#)). Root biomass consequently accumulates within the plough layer, where continued root-derived organic-matter inputs and biopore formation maintain the elevated SOM and porosity observed in the 0–20 cm interval. Reduced soil organic matter content in the plough pan impairs the formation of stable soil aggregates, further decreasing soil porosity and permeability ([Zhao, 2013](#)). This finding aligns with [Bertolino et al. \(2010\)](#), who reported a 44% reduction in soil porosity in plough pans, reflecting a synergistic degradation mechanism driven by reduced porosity and soil organic matter content. The higher clay content in the plough pan results from long-term mechanical compaction and the downward migration of fine particles during rainfall events ([Hua et al., 2008](#)). These differences in soil structure, texture and organic matter content influence hydrological processes, affecting soil moisture availability, infiltration rates, and runoff in agricultural systems ([Horn and Smucker, 2005](#); [Wang and Zhang, 2017](#); [Vereecken et al., 2022](#)). They establish the plough pan as a hydrological barrier in intensively managed wheat–maize systems.

4.2. Soil water management implications

The limited vertical storage capacity of the plough layer renders these soils vulnerable to rapid saturation during intense rainfall. Over the past decade, maximum daily precipitation in the study area ranged from 44.4 mm to 258.3 mm, with a 10-year mean of 114.6 mm ([Fig. 9a](#)), exceeding the plough-layer storage capacity of approximately 39 mm. When precipitation exceeds this storage threshold, soils underlain by a plough pan are prone to prolonged saturation ([Bertolino et al., 2010](#); [Jeřábek et al., 2017](#)), increasing waterlogging risk ([Fig. 9b](#)). The problem is compounded by the low vertical infiltration capacity of the plough pan. In the absence of lateral flow, vertical infiltration rate controlled by plough pan (4.4 mm h^{-1}) falls below 5.2 mm h^{-1} , a threshold for harmful subsoil compaction ([Zink et al., 2011](#)), which classifies the experimental soils as “very dense” under FAO structural

criteria ([FAO, 2015](#)). This restricts drainage and gas exchange, increasing flood risk ([Ma et al., 2024](#)), and decreasing aeration ([Luz et al., 2022](#)). Restoring vertical hydrological connectivity in mechanically compacted soils is therefore a critical strategy for mitigating plough pan-induced waterlogging and improving farmland water management.

Over the longer term, the hydrological impact of the plough pan can be mitigated by restoring vertical hydrological connectivity primarily by increasing the porosity and organic-matter content of the compacted layer. Practical approaches include periodic subsoiling, no-tillage or reduced tillage to avoid repeated compaction at a fixed depth, biological tillage with deep-rooted cover crops, and agroforestry. In the shorter term, irrigation management offers a complementary lever. For wheat–maize rotation systems with a plough pan, reducing the volume of individual irrigation events while increasing their frequency would help avoid short-term waterlogging in the upper soil layer.

4.3. Comparison with other methods

[Yang et al. \(2018\)](#) reported that lateral flow mainly occurred after the soil infiltration reached a steady state. This conclusion differs from the present finding of early-onset lateral infiltration. Yang et al. used repacked soil without macropores in rectangular soil column, so lateral flow could only develop when the wetting front had reached the plough pan and saturated the overlying plough layer. In the present field-based study, by contrast, the undisturbed soil contained macropore channels that conducted water laterally from the initial stage of infiltration. In a contrasting field-based approach, ([Buttle and McDonald, 2002](#)) intercepted subsurface lateral flow at a downslope trench face on a forested hillslope. Lateral infiltration accounted for 59% of the applied irrigation water in their study, lower than the 87% proportion observed in the present study. This discrepancy likely reflects differences in initial conditions. Lateral flow is more readily triggered in wet soils ([Guo et al., 2014](#)). The initial soil water content was 19% in their study, lower than in ours. Using a different methodology, [Wang et al. \(2011\)](#) combined Hydrus–2D simulation with a water balance to quantify lateral infiltration in a peanut monoculture. Their results showed that lateral infiltration accounted for 35–42% of annual rainfall, also lower than the 87% proportion observed here. The key difference here is depth to the impermeable layer (1.3 m at Wang et al.’s site vs. 0.2 m in the present study). The greater thickness of soil above the impermeable layer provides a larger water-storage capacity, retaining more water in the profile and reducing the contribution of lateral flow.

By contrast, [Ehrhardt et al. \(2022\)](#) reported that lateral infiltration accounted for nearly 100% of the total water input which was higher

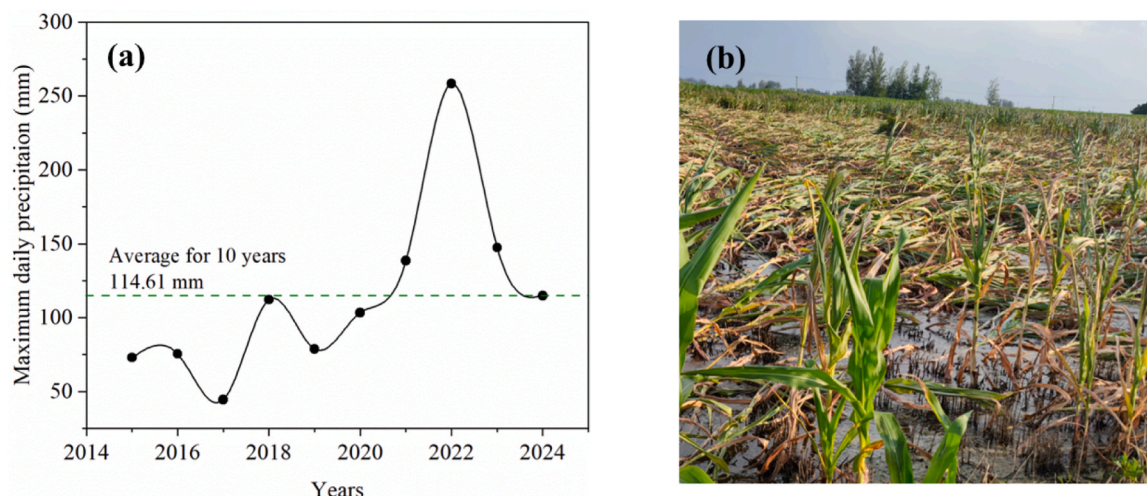


Fig. 9. Maximum daily precipitation in the past 10 years (2014–2024 years) (a), and waterlogging disaster in experimental area in 2024 (b).

than the 87% observed here. In their experimental setup, artificial macropore were constructed in the upper horizon of an undisturbed soil monolith, with the channel bottoms reaching the top of an underlying horizon of lower saturated hydraulic conductivity. Brilliant Blue dye solution was then injected into these macropores, and all outflow was discharged through the lateral tension disc. This higher lateral proportion may be due to the initial soil moisture conditions. The monolith was close to saturation prior to infiltration, leaving the soil matrix with little capacity to absorb water, so water traveled through the macropores to the horizon with lower saturated hydraulic conductivity and was discharged laterally. The method could not be performed under field conditions. To investigate lateral flow directly in the field, [Ehrhardt and Gerke \(2023\)](#) compared soil temperature time series between a lysimeter (vertical infiltration) and the corresponding field soil (vertical and lateral infiltration). Applying wavelet coherence analysis to the temperature differences between the two settings, they verified lateral flow. However, this method could not quantify lateral infiltration. Quantitative analysis would require additional soil moisture data. The present study extends this line of work by quantifying lateral infiltration directly under field conditions, calculating it as the difference between total and vertical infiltration measured with two double-ring infiltrometers at contrasting insertion depths. This dual-depth approach requires no specialized instrumentation beyond standard infiltrometer equipment, making it simple and easy to operate.

4.4. The limitations of the study

This study was carried out at a single site representing one soil type within a wheat–maize rotation system on the Huang-Huai-Hai Plain, where the clay fraction was only 0.6%. In the study site, preferential flow is dominated by structural macropores, including root channels, earthworm burrows, and inter-aggregate voids in the plough layer. In soils with higher clay content, desiccation-induced vertical cracking during wet–dry cycles can create transient macropores that may bypass the plough pan, affecting the validity of the methods. The extension of the dual-depth double-ring approach to clay-rich soils is a priority for our future work. The experiments were also conducted only during the wheat jointing stage in April 2024. Root activity, and antecedent soil moisture differ substantially among periods such as after sowing, after ploughing, and following the rainy season, and the partitioning between vertical and lateral infiltration may vary accordingly. These temporal dynamics will be systematically examined in our subsequent research.

A further methodological limitation is that the depth of the plough pan should be known a priori, so that the deep infiltrometer can be installed with its lower edge resting on, but not penetrating the upper boundary of the pan. For new locations, a preliminary survey, using penetrometer profiling, electrical resistivity tomography ([Jerábek et al., 2017](#)), or exploratory pit excavation, would be required before applying the method. In addition, the dual-depth double-ring methodology proved effective in soils with a well-developed subsurface hydraulic barrier such as a plough pan, but it is not suitable for ecosystems without such barriers (e.g., forests and grasslands), where lateral flow dynamics are governed by fundamentally different mechanisms.

5. Conclusions

This study employed a novel combination of dye tracing and dual-depth double-ring infiltrometers (installed at 2–3 cm and 15–20 cm depths) to quantify vertical and lateral infiltration in plough pan-affected agricultural soils on the Huang-Huai-Hai Plain. The plough pan acted as a hydraulic barrier that redirected most infiltrating water into subsurface lateral flow. Key findings revealed that SLIR accounted for 96% of the steady-state total infiltration rate. LCI accounted for 87% of TCI at a total cumulative water input of 300 mm, with lateral pathways diverting water from vertical percolation. By contrast, vertical infiltration in the absence of lateral infiltration, exhibited severely

limited capacity (SVIR: 4.4 mm h⁻¹; VCI: 40.1 mm at 300 mm cumulative water input), demonstrating the plough pan's role as a hydrological barrier. Post-infiltration soil water storage was concentrated almost entirely within the plough layer, which retained 96% of the total profile storage gain. The limited vertical storage capacity of the plough layer renders these soils vulnerable to rapid saturation and waterlogging during intense rainfall events that are common in the study region. To mitigate waterlogging and improve soil resilience, agricultural management should prioritize strategies that disrupt the plough pan and restore vertical hydrological connectivity. Future flood control frameworks should incorporate plough pan-specific hydraulic thresholds into predictive models to support adaptive water management in pan-dominated agroecosystems. These quantitative results are derived from a silt-loam soil and will be generalized with caution to soils of substantially different texture or structure. Building on the present findings, future research will evaluate whether management practices such as subsoiling or the introduction of deep-rooted cover crops can restore vertical hydrological connectivity through the plough pan and reduce waterlogging risk in long-term tilled farmland.

CRedit authorship contribution statement

Jing Zhang: Writing – original draft, Methodology, Investigation, Formal analysis. **Lingchao Meng:** Writing – original draft, Formal analysis. **Juan Pinos:** Writing – review & editing, Writing – original draft, Formal analysis. **Zhiao Cui:** Investigation. **Manuel Lopez-Vicente:** Writing – review & editing, Writing – original draft, Formal analysis. **Gao-Lin Wu:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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